Inner Product space

- Inner product spaces: Definition and Examples
- The lengths and angles of vectors
- Orthogonal projections
  - Orthogonal complements
- Gram-Schmidt orthogonalization
- Matrix representations of inner products

Learning objectives

- Understand the definition of an inner product space and verify inner product properties in different vector spaces.
- Compute inner products in standard spaces such as $\mathbb{R}^n$ and function spaces.
- Determine vector length (norm), distance between vectors, and angles using inner products.
- Identify and verify orthogonality between vectors and subspaces.
- Apply orthogonal projection to decompose vectors into components parallel and perpendicular to a subspace.
- Determine orthogonal complements and understand their properties in finite-dimensional spaces.
- Construct orthonormal bases using the Gram–Schmidt orthogonalization process.
- Represent inner products using matrices and understand the role of symmetric positive definite matrices.

1 Inner Product space:

Real Inner Product

Definition 1.
Let V be a real vector space. A real inner product on V is a mapping

$$\langle \cdot, \cdot \rangle : V \times V \to \mathbb{R}$$

that assigns to each ordered pair of vectors (α, β) of V to a real number $\langle \alpha, \beta \rangle$.
Satisfying the following properties:
(i) $\langle \alpha, \alpha \rangle > 0$ if $\alpha \neq 0$ (positivity)
(ii) $\langle \alpha, \beta \rangle = \langle \beta, \alpha \rangle$ for all $\alpha, \beta \in V$ (symmetry)
(iii) $\langle \alpha, \beta + \gamma \rangle = \langle \alpha, \beta \rangle + \langle \alpha, \gamma \rangle$ for all $\alpha, \beta, \gamma \in V$ (linearity)
(iv) $\langle c\alpha, \beta \rangle = c\langle \alpha, \beta \rangle = \langle \alpha, c\beta \rangle$ for all $\alpha, \beta \in V,\ c \in \mathbb{R}$

If $\alpha = 0$ then $\langle \alpha, \alpha \rangle = 0$ follows from (iv).

A real vector space V together with a real inner product defined on it is said to be a **Euclidean space**.

Example 1. $V = \mathbb{R}^n$

$x = (a_1, a_2, \ldots, a_n), \quad y = (b_1, b_2, \ldots, b_n)$

$$\langle x, y \rangle = x^T y = \sum_{i=1}^{n} a_i b_i$$

This is called standard inner product.

Example 2. If $\langle x, y \rangle$ is an inner product on a vector space V & $k > 0$, we may define by another inner product by

$$\langle x, y \rangle' = k\langle x, y \rangle$$

Note: If $k \leq 0$, then (i) does not satisfy.

Example 3. Let $V = C[0, 1]$. The vector space of all real valued continuous functions on $[0, 1]$.

For $f, g \in V$ define

$$\langle f, g \rangle = \int_0^1 f(t)g(t)\, dt$$

Note: If $f \neq 0$ then f^2 is bounded away from zero on some subinterval of $[0, 1]$
Hence

$$\langle f, f \rangle = \int_0^1 [f(t)]^2 dt > 0$$

Theorem 1. Let V be an inner product space. Then for $x, y, z \in V$ & $c \in \mathbb{F}$ the followings are true

(i) $\langle x, 0 \rangle = \langle 0, x \rangle = 0$
(ii) $\langle x, x \rangle = 0 \iff x = 0$
(iii) $\langle x, y \rangle = \langle x, z \rangle$ for all $x \in V \Rightarrow y = z$.

Length (Norm) in an Inner Product Space

Let $(V, \langle \cdot, \cdot \rangle)$ be an inner product space. The length (or norm) of a vector $x \in V$ is defined by

$$\|x\| = \sqrt{\langle x, x \rangle}.$$

Angle Between Two Vectors

For non-zero vectors $x, y \in V$, the angle θ between them is defined by

$$\cos\theta = \frac{\langle x, y \rangle}{\|x\| \, \|y\|}, \quad 0 \leq \theta \leq \pi.$$

Equivalently,

$$\theta = \cos^{-1}\left(\frac{\langle x, y \rangle}{\|x\| \, \|y\|} \right).$$

Example 4. Let $V = \mathbb{R}^n$, $x = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$. Then

$$\|x\| = \left\| \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \right\| = \sqrt{a_1^2 + a_2^2 + \cdots + a_n^2}$$

$n = 1$ then $\|a\| = |a|$.

Theorem 2. Let x be a vector in a Euclidean space V and $\|x\|$ be its norm.
Then (i) $\|cx\| = |c|\|x\|$
(ii) $\|x\| > 0$ when $x \neq 0$. When $x = 0$ then $\|0\| = 0$

Proof.

$$\|cx\| = \sqrt{\langle cx, cx \rangle} = \sqrt{c^2 \langle x, x \rangle}$$
$$= |c| \sqrt{\langle x, x \rangle} = |c| \|x\|$$

□

Schwartz's inequality

For any two vector x, y in V (Euclidean space)

$$|\langle x, y \rangle| \leq \|x\| \|y\|$$

where the equality holds when x & y are linearly dependent.

Proof. • **Case-1:** Let x & y or both x and y be null vectors.

Then both sides being zero equality holds.

- **Case-2:**

  Let x & y be non-null linearly independent vectors.

  Then there exists a real number k such that

  $$x = ky$$

  Then $\|x\| = |k|\|y\|$

  $$\langle x, y\rangle = \langle ky, y\rangle = k\langle y, y\rangle = k\|y\|^2$$

  Therefore

  $$|\langle x, y\rangle| = |k|\|y\|^2 = \|x\|\|y\|$$

- **Case-3:** Let x & y be not linearly dependent.

  Then $x - ky \neq 0$ for all real number k

  $$\Rightarrow \langle x - ky, x - ky\rangle > 0$$
  $$\Rightarrow \langle x, x\rangle - 2k\langle x, y\rangle + k^2\langle y, y\rangle > 0$$

  Since $\langle x, x\rangle$, $\langle x, y\rangle$ & $\langle y, y\rangle$ are real & $\langle y, y\rangle \neq 0$

  Then LHS is a real quadratic polynomial in k which is positive for all real value k, the discriminant of the quadratic polynomial must be negative. Otherwise the polynomial would be zero for some real k.

  Thus,

  $$\langle x, y\rangle^2 - \langle x, x\rangle\langle y, y\rangle < 0$$
  $$\Rightarrow \langle x, y\rangle^2 < \langle x, x\rangle\langle y, y\rangle$$
  $$\Rightarrow |\langle x, y\rangle| \leq \|x\|\|y\|$$

□

Triangle inequality

For any two vectors x, y in a Euclidean space V then

$$\|x + y\| \leq \|x\| + \|y\|$$

Proof.

$$\|x + y\|^2 = \langle x + y, x + y\rangle = \langle x, x\rangle + 2\langle x, y\rangle + \langle y, y\rangle$$
$$\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\| + \|y\|)^2$$
$$\Rightarrow \|x + y\| \leq \|x\| + \|y\|$$

□

Note 1.

- If $\|x + y\| = \|x\| + \|y\|$ then x & y are linearly dependent.

- $\|x+y\| = \|x\| + \|y\| \Rightarrow$ L.D but converse is not true. $x = (1,2,3) \quad y = (-1,-2,-3)$

  x & y are L.D but $\|x+y\| \neq 0$

  $\|x+y\| = 0$ but $\|x\| \neq 0, \|y\| \neq 0$

A vector x in an Euclidean space V is said to be **unit vector** if $\|x\| = 1$
If x is a non zero vector in V then $\frac{x}{\|x\|}$ is a unit vector.
In a Euclidean space x is said to be **orthogonal** to y if $\langle x, y\rangle = 0$ i.e. $\langle x, y\rangle = 0 \Rightarrow x \perp y$
If x is orthogonal to y i.e. $\langle x, y\rangle = 0$ then x is also orthogonal to x
The null vector 0 is orthogonal to any non zero vector and also orthogonal to itself.

Pythagoras theorem

If x & y be two orthogonal vectors in a Euclidean space V, then

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2$$

Proof.

$$\begin{aligned}\|x+y\|^2 &= \langle x+y, x+y\rangle \\ &= \langle x, x\rangle + 2\langle x, y\rangle + \langle y, y\rangle \\ &= \|x\|^2 + \|y\|^2 \quad \text{as } \langle x, y\rangle = 0\end{aligned}$$

□

Parallelogram Law

If x, y be any two vectors in a Euclidean space V then

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

Proof.

$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 + 2\langle x, y\rangle$$
$$\|x-y\|^2 = \|x\|^2 + \|y\|^2 - 2\langle x, y\rangle$$

Adding,

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

□

The vectors $(x-y)$ and $(x+y)$ corresponds to the diagonal of the parallelogram. The sum of the squares of the lengths of the diagonals is $\|x+y\|^2 + \|x-y\|^2$. The sum of the squares of the lengths of the sides is $2(\|x\|^2 + \|y\|^2)$.

Orthogonal set of vectors

A set of vectors $\{\beta_1, \beta_2, \ldots, \beta_n\}$ in an Euclidean space is said to be orthogonal if

$$\langle \beta_i, \beta_j \rangle = 0 \quad \text{whenever } i \neq j.$$

A set of vectors $\{\beta_1, \beta_2, \ldots, \beta_n\}$ is said to be orthonormal if

$$\langle \beta_i, \beta_j \rangle = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

Note 2. An orthogonal set may contain the zero vector, but an orthonormal set cannot contain the zero vector.

Theorem 3. An orthogonal set of non-null vectors in a Euclidean space V is linearly independent.

Proof. Let $\{\beta_1, \beta_2, \ldots, \beta_n\}$ be an orthogonal set of non-null vectors.

Let

$$c_1\beta_1 + c_2\beta_2 + \cdots + c_n\beta_n = 0.$$

Taking inner product with β_i:

$$\langle c_1\beta_1 + c_2\beta_2 + \cdots + c_n\beta_n, \beta_i \rangle = 0$$

$$\Rightarrow c_1\langle \beta_1, \beta_i \rangle + \cdots + c_n\langle \beta_n, \beta_i \rangle = 0$$

$$\Rightarrow c_i\langle \beta_i, \beta_i \rangle = 0$$

$$\Rightarrow c_i\|\beta_i\|^2 = 0$$

$$\Rightarrow c_i = 0 \quad (\text{since } \beta_i \neq 0)$$

Hence all $c_i = 0 \Rightarrow \{\beta_1, \ldots, \beta_n\}$ is linearly independent. □

Example 5. In $\mathbb{R}^3$, $\{(1,1,0), (1,-1,0), (-1,1,2)\}$ is an orthogonal set of non-zero vectors, but it is not orthonormal. However, we can normalize the vectors to obtain an orthonormal set:

$$\left\{ \frac{1}{\sqrt{2}}(1,1,0),\ \frac{1}{\sqrt{2}}(1,-1,0),\ \frac{1}{\sqrt{6}}(-1,1,2) \right\}$$

2 Orthogonal Projection on a Subspace

If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$ is an orthonormal basis for subspace W of an inner product space V and $\mathbf{w}$ is any vector in V, then the orthogonal projection of $\mathbf{w}$ on W is given by

$$\text{proj}_W \mathbf{w} = \langle \mathbf{w}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{w}, \mathbf{v}_2 \rangle \mathbf{v}_2 + \cdots + \langle \mathbf{w}, \mathbf{v}_r \rangle \mathbf{v}_r.$$

Note 3. If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_r\}$ is an orthogonal basis for W and $\mathbf{w}$ is any vector in V, then

$$\text{proj}_W \mathbf{w} = \frac{\langle \mathbf{w}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2} \mathbf{v}_1 + \frac{\langle \mathbf{w}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2} \mathbf{v}_2 + \cdots + \frac{\langle \mathbf{w}, \mathbf{v}_r \rangle}{\|\mathbf{v}_r\|^2} \mathbf{v}_r.$$

2.1 Projection Theorem

Theorem 4.8: If W is a subspace of a finite dimensional inner product space V and $\mathbf{w}$ is any vector in V, then

$$\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$$

where

$$\mathbf{w}_1 = \text{proj}_W \mathbf{w} \quad \text{is in } W$$

and

$$\mathbf{w}_2 = \text{proj}_{W^\perp} \mathbf{w} \quad \text{is in } W^\perp.$$

Example 6. Find the orthogonal projection of $\mathbf{u} = (1, -2, 3)$ along $\mathbf{v} = (1, 2, 1)$ in $\mathbb{R}^3$ with respect to the Euclidean inner product.

Solution: Let W be the subspace spanned by the vector $\mathbf{v} = (1, 2, 1)$.

$$\text{proj}_W \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}\mathbf{v} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}$$

$$= \frac{(1, -2, 3) \cdot (1, 2, 1)}{(1^2 + 2^2 + 1^2)}(1, 2, 1) = \frac{1 - 4 + 3}{6}(1, 2, 1) = \frac{0}{6}(1, 2, 1) = \mathbf{0}$$

Example 7. Find the orthogonal projection of $\mathbf{u} = (4, 0, -1)$ along $\mathbf{v} = (3, 1, -5)$ in $\mathbb{R}^3$ with respect to the Euclidean inner product. Also find the component of $\mathbf{u}$ orthogonal to $\mathbf{v}$.

Solution: Let W be the subspace spanned by the vector $\mathbf{v} = (3, 1, -5)$.

$$\text{proj}_W \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{v}\|^2}\mathbf{v} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}$$

$$= \frac{(4, 0, -1) \cdot (3, 1, -5)}{(9 + 1 + 25)}(3, 1, -5) = \frac{17}{35}(3, 1, -5) = \left(\frac{51}{35}, \frac{17}{35}, -\frac{17}{7}\right)$$

The component of $\mathbf{u}$ orthogonal to $\mathbf{v}$ is

$$\text{proj}_{W^\perp} \mathbf{u} = \mathbf{u} - \text{proj}_W \mathbf{u} = (4, 0, -1) - \left(\frac{51}{35}, \frac{17}{35}, -\frac{17}{7}\right) = \left(\frac{89}{35}, -\frac{17}{35}, \frac{10}{7}\right)$$

Example 8. Let W be the subspace of $\mathbb{R}^3$ with orthonormal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$, where

$$\mathbf{v}_1 = \left(\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right), \quad \mathbf{v}_2 = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right).$$

Using Euclidean inner product find the distance from the vector $\mathbf{u} = (1, 1, 0)$ to W, where $\mathbf{u}$ is in V.

Solution: From the approximation theorem,

$$\text{Distance from } \mathbf{u} \text{ to } W = \|\mathbf{u} - \text{proj}_W \mathbf{u}\|$$

$$\text{proj}_W \mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2$$

$$= \left\{(1, 1, 0) \cdot \left(\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right)\right\}\left(\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right) + \left\{(1, 1, 0) \cdot \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)\right\}\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$

$$= \frac{1}{3}\left(\frac{2}{3}, -\frac{1}{3}, -\frac{2}{3}\right) + \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right) = \left(\frac{13}{18}, -\frac{1}{9}, \frac{5}{18}\right)$$

$$\|\mathbf{u} - \text{proj}_W \mathbf{u}\| = \left\|(1,1,0) - \left(\frac{13}{18}, -\frac{1}{9}, \frac{5}{18}\right)\right\| = \left\|\left(\frac{5}{18}, -\frac{10}{9}, -\frac{5}{18}\right)\right\|$$

$$= \sqrt{\frac{25}{324} + \frac{100}{81} + \frac{25}{324}} = \frac{5\sqrt{2}}{6}$$

Example 9. Let W be the subspace spanned by the orthonormal vectors

$$\mathbf{v}_1 = (0,1,0) \quad \text{and} \quad \mathbf{v}_2 = \left(\frac{4}{5}, 0, -\frac{3}{5}\right).$$

Find the orthogonal projection of $\mathbf{w} = (1,2,3)$ on W with respect to Euclidean inner product. Also find the component of $\mathbf{w}$ orthogonal to W and express $\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2$ where $\mathbf{w}_1$ is in W and $\mathbf{w}_2$ is in $W^\perp$.

Solution: Since an orthonormal set is also linearly independent, the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ form an orthonormal basis for W.

$$\text{proj}_W \mathbf{w} = \langle \mathbf{w}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{w}, \mathbf{v}_2 \rangle \mathbf{v}_2$$

$$= (1,2,3) \cdot (0,1,0)(0,1,0) + (1,2,3) \cdot \left(\frac{4}{5}, 0, -\frac{3}{5}\right)\left(\frac{4}{5}, 0, -\frac{3}{5}\right)$$

$$= (0,2,0) + (-1)\left(\frac{4}{5}, 0, -\frac{3}{5}\right) = \left(-\frac{4}{5}, 2, \frac{3}{5}\right)$$

The component of $\mathbf{w}$ orthogonal to W is

$$\text{proj}_{W^\perp} \mathbf{w} = \mathbf{w} - \text{proj}_W \mathbf{w} = (1,2,3) - \left(-\frac{4}{5}, 2, \frac{3}{5}\right) = \left(\frac{9}{5}, 0, \frac{12}{5}\right)$$

Using Projection theorem $\mathbf{w}$ can be expressed as

$$\mathbf{w} = \mathbf{w}_1 + \mathbf{w}_2 = \text{proj}_W \mathbf{w} + \text{proj}_{W^\perp} \mathbf{w} = \left(-\frac{4}{5}, 2, \frac{3}{5}\right) + \left(\frac{9}{5}, 0, \frac{12}{5}\right)$$

2.2 Additional Examples on Orthogonal Projection

Example 10. Let W be the subspace of $\mathbb{R}^3$ spanned by $\mathbf{v} = (1,2,3)$. Find the orthogonal projection of $\mathbf{u} = (4,5,6)$ onto W and the component of $\mathbf{u}$ orthogonal to W.

Solution:

$$\text{proj}_W \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2}\mathbf{v}$$

$$\mathbf{u} \cdot \mathbf{v} = 4 + 10 + 18 = 32, \quad \|\mathbf{v}\|^2 = 1 + 4 + 9 = 14$$

$$\text{proj}_W \mathbf{u} = \frac{32}{14}(1,2,3) = \frac{16}{7}(1,2,3) = \left(\frac{16}{7}, \frac{32}{7}, \frac{48}{7}\right)$$

Component of $\mathbf{u}$ orthogonal to W:

$$\mathbf{u} - \text{proj}_W \mathbf{u} = (4,5,6) - \left(\frac{16}{7}, \frac{32}{7}, \frac{48}{7}\right) = \left(\frac{12}{7}, \frac{3}{7}, -\frac{6}{7}\right)$$

Example 11. Find the orthogonal projection of $\mathbf{u} = (3,-1,4)$ onto the subspace W of $\mathbb{R}^3$ spanned by the orthogonal vectors $\mathbf{v}_1 = (1,1,1)$ and $\mathbf{v}_2 = (1,-1,0)$.

Solution:

$$\text{proj}_W \mathbf{u} = \frac{\langle \mathbf{u}, \mathbf{v}_1 \rangle}{\|\mathbf{v}_1\|^2}\mathbf{v}_1 + \frac{\langle \mathbf{u}, \mathbf{v}_2 \rangle}{\|\mathbf{v}_2\|^2}\mathbf{v}_2$$

$$\langle \mathbf{u}, \mathbf{v}_1 \rangle = 6, \quad \|\mathbf{v}_1\|^2 = 3 \implies 2(1,1,1) = (2,2,2)$$

$$\langle \mathbf{u}, \mathbf{v}_2 \rangle = 4, \quad \|\mathbf{v}_2\|^2 = 2 \implies 2(1,-1,0) = (2,-2,0)$$

$$\text{proj}_W \mathbf{u} = (2,2,2) + (2,-2,0) = (4,0,2)$$

Example 12. Let W be the subspace of $\mathbb{R}^4$ with orthonormal basis

$$\mathbf{v}_1 = \left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right), \quad \mathbf{v}_2 = \left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0, 0\right).$$

Find the orthogonal projection of $\mathbf{u} = (1,2,3,4)$ onto W.

Solution:

$$\text{proj}_W \mathbf{u} = \langle \mathbf{u}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \langle \mathbf{u}, \mathbf{v}_2 \rangle \mathbf{v}_2 = \left(2, 3, \frac{5}{2}, \frac{5}{2}\right)$$

Example 13. Find the distance from the point $\mathbf{u} = (5,1,4)$ to the plane W in $\mathbb{R}^3$ given by the equation $x + 2y - z = 0$.

Solution:
Normal vector $\mathbf{n} = (1,2,-1)$.

$$\text{proj}_W \mathbf{u} = \frac{3}{6}(1,2,-1) = \left(\frac{1}{2}, 1, -\frac{1}{2}\right)$$

Distance $= \|\mathbf{u} - \text{proj}_W \mathbf{u}\| = \dfrac{9\sqrt{2}}{2}$

Example 14. Let $W = \text{span}\{(1,0,1),(0,1,1)\}$ in $\mathbb{R}^3$. Find the orthogonal decomposition of $w = (3,4,5)$ as $w = w_1 + w_2$, where $w_1 \in W$ and $w_2 \in W^{\perp}$.

Solution:
Let $v_1 = (1,0,1)$ and $v_2 = (0,1,1)$. Assume

$$w_1 = av_1 + bv_2 = (a, b, a+b)$$

Then

$$w_2 = w - w_1 = (3-a,\ 4-b,\ 5-(a+b))$$

Since $w_2 \in W^\perp$, we must have:

$$w_2 \cdot v_1 = 0 \quad \text{and} \quad w_2 \cdot v_2 = 0$$

Compute:

$$(3-a,\, 4-b,\, 5-(a+b)) \cdot (1,0,1) = (3-a) + (5-(a+b)) = 8 - 2a - b = 0$$

$$(3-a,\, 4-b,\, 5-(a+b)) \cdot (0,1,1) = (4-b) + (5-(a+b)) = 9 - a - 2b = 0$$

Solve the system:

$$8 - 2a - b = 0, \quad 9 - a - 2b = 0$$

From the first equation, $b = 8 - 2a$. Substituting into the second:

$$9 - a - 2(8 - 2a) = 0 \Rightarrow 9 - a - 16 + 4a = 0 \Rightarrow 3a - 7 = 0$$

$$a = \frac{7}{3}, \quad b = \frac{10}{3}$$

Thus,

$$w_1 = \left(\frac{7}{3}, \frac{10}{3}, \frac{17}{3}\right)$$

$$w_2 = w - w_1 = \left(\frac{2}{3}, \frac{2}{3}, -\frac{2}{3}\right)$$

$$\boxed{w_1 = \left(\frac{7}{3}, \frac{10}{3}, \frac{17}{3}\right), \quad w_2 = \left(\frac{2}{3}, \frac{2}{3}, -\frac{2}{3}\right)}$$

2.3 Orthogonal complement:

Let W be a subspace of $\mathbb{R}^n$. Its orthogonal complement is

$$W^\perp = \{v \in \mathbb{R}^n \mid \langle v, w \rangle = 0 \ \forall w \in W\}.$$

Example: The orthogonal complement of $\mathbb{R}^n$ is $\{0\}$.
Same reason, $\{0\}^\perp = \mathbb{R}^n$.
Computing orthogonal complement:
Let A be a matrix and $W = \text{Col}(A)$.
Then

$$W^\perp = \text{Null}(A^T).$$

Example: Compute $W^\perp$ where

$$W = \text{span}\left\{\begin{pmatrix}1\\2\\1\end{pmatrix}, \begin{pmatrix}-3\\2\\1\end{pmatrix}\right\}.$$

$$A = \begin{pmatrix}1 & -3\\2 & 2\\1 & 1\end{pmatrix}$$

$$A^T = \begin{pmatrix} 1 & 2 & 1 \\ -3 & 2 & 1 \end{pmatrix}$$

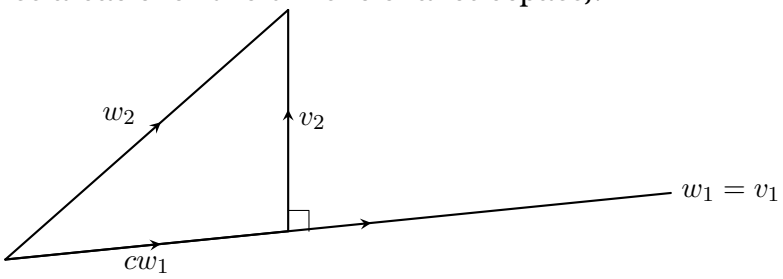
$$A^T x = 0 \Rightarrow \begin{cases} x_1 + 2x_2 + x_3 = 0 \\ -3x_1 + 2x_2 + x_3 = 0 \end{cases}$$

$$\Rightarrow x_1 = \frac{7x_2}{2}, \quad x_3 = -\frac{17x_2}{2}$$

$$\Rightarrow x = t \begin{pmatrix} 7 \\ 2 \\ -17 \end{pmatrix}$$

3 Gram-Schmidt Orthogonalization Process

Suppose that $\{w_1, w_2\}$ is a linearly independent subset of an inner product space (and hence a basis for two dimensional subspace).



We want to construct an orthogonal set from $\{w_1, w_2\}$ that spans the same subspace.

Let

$$v_1 = w_1, \quad v_2 = w_2 - cw_1$$

From the figure, $v_1 = w_1$ and $v_2 = w_2 - cw_1$ has this property.

Choose c so that v_2 is orthogonal to v_1:

$$0 = \langle v_2, v_1 \rangle = \langle w_2 - cw_1, w_1 \rangle$$

$$= \langle w_2, w_1 \rangle - c\langle w_1, w_1 \rangle$$

$$\Rightarrow c = \frac{\langle w_2, w_1 \rangle}{\langle w_1, w_1 \rangle}$$

Hence $v_1 = w_1$ and

$$v_2 = w_2 - \frac{\langle w_2, w_1 \rangle}{\langle w_1, w_1 \rangle} w_1$$

which impilies that $v_1 \perp v_2$.

Note: cw_1 is called projection of w_2 along w_1. $\langle w_2, w_1 \rangle / \|w_1\|$ is called scalar component of w_2 upon w_1.

3.1 Motivation of the Gram–Schmidt Process

In many problems of linear algebra, we start with a set of linearly independent vectors

$$\{v_1, v_2, \ldots, v_n\}$$

which forms a basis for a vector space. However, these vectors may not be orthogonal to each other. Working with a non-orthogonal basis is often difficult, especially when we want to compute projections, coordinates, distances, or least-squares approximations.

The main motivation of the Gram–Schmidt process is to construct a new set of vectors

$$\{u_1, u_2, \ldots, u_n\}$$

from the given vectors such that

$$\operatorname{span}\{u_1, u_2, \ldots, u_n\} = \operatorname{span}\{v_1, v_2, \ldots, v_n\},$$

but the new vectors are mutually orthogonal, that is,

$$u_i \cdot u_j = 0, \qquad i \neq j.$$

Thus, the Gram–Schmidt process does not change the vector space generated by the original vectors; it only replaces the given basis by a more convenient orthogonal basis.

The basic idea is to remove from each vector the component that lies in the directions of the previously constructed orthogonal vectors. For example, if

$$u_1 = v_1,$$

then the part of v_2 in the direction of u_1 is

$$\operatorname{proj}_{u_1}(v_2) = \frac{v_2 \cdot u_1}{u_1 \cdot u_1} u_1.$$

Therefore, we define

$$u_2 = v_2 - \operatorname{proj}_{u_1}(v_2).$$

This makes u_2 orthogonal to u_1.

In general, the Gram–Schmidt process gives

$$u_k = v_k - \sum_{j=1}^{k-1} \operatorname{proj}_{u_j}(v_k),$$

where

$$\operatorname{proj}_{u_j}(v_k) = \frac{v_k \cdot u_j}{u_j \cdot u_j} u_j.$$

If required, each orthogonal vector can be normalized by defining

$$e_k = \frac{u_k}{\|u_k\|}.$$

Then the set

$$\{e_1, e_2, \ldots, e_n\}$$

becomes an orthonormal basis.

Hence, the Gram–Schmidt process is important because it transforms any linearly independent set into an orthogonal or orthonormal basis, which makes many computations in linear algebra simpler and more efficient.

3.2 Gram Schmidt Process

The **Gram Schmidt Process** is used to convert a linearly independent set of vectors $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_k\}$ into an **orthogonal** or **orthonormal** basis for the same subspace.

Let V be an inner product space and let $S = \{w_1, w_2, \ldots, w_n\}$ be a linearly independent subset of V.

Define: $S' = \{v_1, v_2, \ldots, v_n\}$

$$v_1 = w_1$$

$$v_2 = w_2 - \frac{\langle w_2, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1$$

$$v_3 = w_3 - \frac{\langle w_3, v_1 \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle w_3, v_2 \rangle}{\langle v_2, v_2 \rangle} v_2$$

$$\vdots$$

$$v_k = w_k - \sum_{i=1}^{k-1} \frac{\langle w_k, v_i \rangle}{\langle v_i, v_i \rangle} v_i \quad (2 \leq k \leq n)$$

Then $S' = \{v_1, v_2, \ldots, v_n\}$ is an orthogonal set of non-zero vectors such that

$$\text{span}(S) = \text{span}(S')$$

3.2.1 Steps of Gram-Schmidt Process

Let $\{\mathbf{w}_1, \mathbf{w}_2, \ldots, \mathbf{w}_n\}$ be a linearly independent set.

1. Set $\mathbf{v}_1 = \mathbf{w}_1$.
2. For $k = 2, 3, \ldots, n$,
$$\mathbf{v}_k = \mathbf{w}_k - \sum_{i=1}^{k-1} \text{proj}_{\mathbf{v}_i} \mathbf{w}_j$$
where
$$\text{proj}_{\mathbf{v}_i} \mathbf{w}_j = \frac{\langle \mathbf{w}_k, \mathbf{w}_i \rangle}{\|\mathbf{v}_i\|^2} \mathbf{v}_i$$
3. (Optional) To obtain an **orthonormal basis** $\{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_n\}$, normalize each vector:
$$\mathbf{q}_k = \frac{\mathbf{v}_k}{\|\mathbf{v}_k\|}, \quad k = 1, 2, \ldots, n.$$

Example. Use Gram–Schmidt process to obtain an orthonormal basis form the basis set $\{(1, 0, 1), (1, 3, -1), (1, 3, 4)\}$ of the Euclidean space $\mathbb{R}^3$ with standard inner product.

Answer: Let $w_1 = (1, 0, 1), w_2 = (1, 1, 1)$ and $w_3 = (1, 3, 4)$.

Let

$$v_1 = (1, 0, 1)$$

$$v_2 = w_2 - c_1 v_1 \quad \text{where} \quad c_1 = \frac{\langle w_2, v_1 \rangle}{\langle v_1, v_1 \rangle}$$

$$\Rightarrow v_2 = w_2 - \frac{\langle w_2, v_1\rangle}{\langle v_1, v_1\rangle} v_1 = (1,1,1) - \frac{2}{2}(1,0,1) = (0,1,0)$$

as

$$\langle w_2, v_1\rangle = (1)(1) + (0)(1) + (1)(1) = 1 + 1 = 2$$

and

$$\langle v_2, v_1\rangle = (1)(1) + (0)(0) + (1)(1) = 1 + 1 = 2$$

$$v_3 = w_3 - d_1 v_1 - d_2 v_2$$

$$d_1 = \frac{\langle w_3, v_1\rangle}{\langle v_1, v_1\rangle} = \frac{1+0+4}{2} = \frac{5}{2}$$

$$d_2 = \frac{\langle w_3, v_2\rangle}{\langle v_2, v_2\rangle} = \frac{0+3+0}{0+1+0} = 3$$

$$v_3 = (1,3,4) - \frac{5}{2}(1,0,1) - 3(1,3,-1) = \frac{3}{2}(-1,0,1)$$

Thus an orthogonal basis is

$$\{(1,0,1), (0,1,0), \frac{3}{2}(-1,0,1)\}$$

Orthonormal basis is

$$\left\{ \frac{1}{\sqrt{2}}(1,0,1),\ (0,1,0),\ \frac{\sqrt{2}}{3}(-1,0,1) \right\}$$

Example 15. Let $V = P(\mathbb{R})$ with the inner product

$$\langle f(x), g(x)\rangle = \int_{-1}^{1} f(t)g(t)\,dt,$$

and consider the subspace $P_2(\mathbb{R})$ with the standard ordered basis

$$\beta = \{1, x, x^2\}.$$

Use the Gram–Schmidt process to replace β by an orthogonal basis $\{v_1, v_2, v_3\}$ for $P_2(\mathbb{R})$, and then obtain an orthonormal basis.

Solution:

Step 1:

$$v_1 = 1, \quad \|v_1\|^2 = \int_{-1}^{1} 1^2\,dt = 2$$

Step 2:

$$v_2 = x - \frac{\langle x, v_1\rangle}{\|v_1\|^2} v_1$$

$$\langle x, 1\rangle = \int_{-1}^{1} t\,dt = 0$$

$$\Rightarrow v_2 = x, \quad \|v_2\|^2 = \int_{-1}^{1} t^2\,dt = \frac{2}{3}$$

Step 3:

$$v_3 = x^2 - \frac{\langle x^2, v_1\rangle}{\|v_1\|^2}v_1 - \frac{\langle x^2, v_2\rangle}{\|v_2\|^2}v_2$$

$$\langle x^2, 1\rangle = \int_{-1}^{1} t^2\,dt = \frac{2}{3}, \quad \langle x^2, x\rangle = \int_{-1}^{1} t^3\,dt = 0$$

$$\Rightarrow v_3 = x^2 - \frac{1}{3}$$

Orthogonal basis:

$$\{1,\ x,\ x^2 - \tfrac{1}{3}\}$$

Normalization:

$$\|v_1\| = \sqrt{2}, \quad \|v_2\| = \sqrt{\frac{2}{3}}, \quad \|v_3\| = \sqrt{\frac{8}{45}}$$

$$e_1 = \frac{1}{\sqrt{2}}, \quad e_2 = \sqrt{\frac{3}{2}}\,x, \quad e_3 = \sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right)$$

Orthonormal basis:

$$\left\{\frac{1}{\sqrt{2}},\ \sqrt{\frac{3}{2}}\,x,\ \sqrt{\frac{45}{8}}\left(x^2 - \frac{1}{3}\right)\right\}$$

4 Matrix representation of inner product

Let $V = \mathbb{R}^2$ and the inner product is defined by

$$\langle u, v\rangle = 2u_1v_1 + 3u_1v_2 + 3u_2v_1 + 4u_2v_2$$

Have standard basis of V is

$$\{(1,0),(0,1)\}$$

Let $e_1 = (1,0)$ and $e_2 = (0,1)$

The matrix corresponding to the inner product is

$$G = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = \begin{bmatrix} \langle e_1, e_1\rangle & \langle e_1, e_2\rangle \\ \langle e_2, e_1\rangle & \langle e_2, e_2\rangle \end{bmatrix}$$

where

$$a_{11} = \langle e_1, e_1\rangle = 2, \quad a_{12} = \langle e_1, e_2\rangle = 3$$

$$a_{21} = \langle e_2, e_1\rangle = 3, \quad a_{22} = \langle e_2, e_2\rangle = 4$$

$$G = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$$

Exercises

1. In $\mathbb{R}^3$, let $\alpha = (a_1, a_2, a_3)$, $\beta = (b_1, b_2, b_3)$. Determine whether $(\ ,\ )$ is a real inner product for $\mathbb{R}^3$ if $(\ ,\ )$ be defined by

   (a) $(\alpha, \beta) = (a_1 + a_2 + a_3)(b_1 + b_2 + b_3)$;

   (b) $(\alpha, \beta) = a_1 b_1 + (a_2 + a_3)(b_2 + b_3) + a_3 b_3$;

2. Prove that for all α, β in a Euclidean space V

   (a) $(\alpha, \beta) = 0$ if and only if $\|\alpha + \beta\| = \|\alpha - \beta\|$;

   (b) $(\alpha, \beta) = 0$ if and only if $\|\alpha + \beta\|^2 = \|\alpha\|^2 + \|\beta\|^2$.

3. Prove that the set of vectors $\{(2, 3, -1), (1, -2, -4), (2, -1, 1)\}$ is an orthogonal basis of the Euclidean space $\mathbb{R}^3$ with standard inner product. Find the projections of the vector $\alpha = (1, 1, 1)$ along these basis vectors and verify that α is the sum of its projections along these basis vectors.

4. Prove that the set of vectors $\{(1, 0, 0, 1), (-1, 0, 2, 1), (1, 3, 1, -1), (1, -1, 1, -1)\}$ is an orthogonal basis of the Euclidean space $\mathbb{R}^4$ with standard inner product. Find the projections of the vector $\alpha = (1, 2, 3, 1)$ along these basis vectors and express α as a linear combination of these basis vectors.

5. Use Gram-Schmidt process to convert the given basis B of the Euclidean space $\mathbb{R}^3$ with standard inner product into an orthogonal basis.

   (a) $B = \{(1, 2, -2), (2, 0, 1), (1, 1, 0)\}$;

   (b) $B = \{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$.

6. Apply Gram-Schmidt process to find an orthonormal basis for the Euclidean space $\mathbb{R}^3$ with standard inner product, that contains

   (a) the vector $\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right)$;

   (b) the vectors $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right), \left(\frac{2}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$.

7. Apply Gram-Schmidt process to obtain an orthonormal basis of the subspace of the Euclidean space $\mathbb{R}^4$ with standard inner product, spanned by the vectors

   (a) $(1, 1, 0, 1), (1, -2, 0, 0), (1, 0, -1, 2)$;

   (b) $(1, 1, 1, 1), (1, 1, -1, -1), (1, 2, 0, 2)$.

8. Find the orthogonal complement of the row space of the matrix:

   (a) $\begin{pmatrix} 1 & 2 & 1 & 0 \\ 1 & 3 & 2 & 2 \\ 0 & 1 & 1 & 2 \end{pmatrix}$,

   [Hint.] Consider the system of linear equations $AX = 0$ with the given matrix A as coefficient matrix. The solution space of the homogeneous system is the orthogonal complement of the row space of A.

5 Solution

1. (a) No, (b) Yes
2. NA
3. $\frac{2}{7}(2,3,1), -\frac{5}{21}(1,-2,4), \frac{1}{3}(2,-1,1)$
4. $(1,0,0,1), (-1,0,2,1), \frac{3}{4}(1,3,1,-1), \frac{1}{4}(1,-1,1,-1)$
5. (a) $\{(1,2,-2), (2,0,1), \frac{1}{15}(-2,5,4)\}$, (b) $\{(1,1,0), \frac{1}{2}(-1,-1,2), \frac{1}{3}(2,-2,2)\}$
6. (a) $\left\{\left(\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0\right), \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), (0,0,1)\right\}$, (b) $\left\{\frac{1}{\sqrt{3}}(1,1,-1), \frac{1}{\sqrt{6}}(2,-1,1), \frac{1}{\sqrt{2}}(0,1,1)\right\}$
7. (a) $\left\{\frac{1}{\sqrt{3}}(1,1,0,1), \frac{1}{\sqrt{42}}(4,-5,0,1), \frac{1}{\sqrt{105}}(4,2,7,-6)\right\}$,
   (b) $\left\{\frac{1}{4}(1,1,1,1), \frac{1}{4}(1,1,-1,-1), \frac{1}{\sqrt{10}}(-1,1,-2,2)\right\}$
8. $L\{(1,-1,1,0), (4,-2,0,1)\}$